Mathematics - Undergraduate Level Assessment 23 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. Statement 1 | Every ideal in a ring is a subring of the ring. Statement 2 | Every subring of every ring is an ideal of the ring.
   (a) True, True
   (b) False, False
   (c) True, False
   (d) False, True
2. In the xy-plane, the graph of $x^{(}log$ y$) = y^{l}og$ x is
   (a) the open first quadrant
   (b) a closed curve
   (c) a ray in the open first quadrant
   (d) a single point
3. Find the degree for the given field extension Q(sqrt(2) + sqrt(3)) over Q.
   (a) 0
   (b) 4
   (c) 2
   (d) 6
4. Find the generator for the finite field Z_7.
   (a) 1
   (b) 2
   (c) 3
   (d) 4
5. Which of the following is true?
   (a) Every compact space is complete
   (b) Every complete space is compact
   (c) Neither (a) nor (b).
   (d) Both (a) and (b).

True / False (10 marks)

Answer True or False

6. True or False: The characteristic of the ring 2 Z is 12, suggesting it can be generated by twelve elements.

7. True or False: The linear transformation T maps the point (2, 1) to (1, 6).

8. True or False: The equation of the tangent line to the graph of y = x + e^x at x = 0 is y = 2 x + 1.

9. True or False: A complete graph with 10 vertices contains 20 edges, as each vertex connects only to a limited number of others.

10. True or False: Every permutation is a one-to-one function, mapping each element of a set uniquely to another element.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Solve
$$\frac{2x+4}{x^2+4x-5} = \frac{2-x}{x-1}$$
for x.

12. Discuss the implications of Statement 1 regarding the uniform continuity of a continuous function defined on a compact set, and analyze the conditions under which the composition of two differentiable functions is differentiable as stated in Statement 2.

End of Paper